(i)

$$1 = \int_0^1 kx^n(1-x)\,dx = k\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$$
$$\implies k = (n+1)(n+2)$$

$$\begin{aligned}\mu &= \int_0^1 xkx^n(1-x)\,dx \\ &= k\left(\frac{1}{n+2} - \frac{1}{n+3}\right) \\ &= \frac{n+1}{n+3}\end{aligned}$$

(ii)

$$\begin{aligned}F(x) &= \int_0^x (n+1)(n+2)t^n(1-t)\,dt \\ &= (n+2)x^{n+1} - (n+1)x^{n+2} \\ &= x^{n+1}\left(n+2-(n+1)x\right)\end{aligned}$$

$$\begin{aligned}F(\mu) &= \left(\frac{n+1}{n+3}\right)^{n+1}\left(n+2-(n+1)\frac{n+1}{n+3}\right) \\ &= \left(\frac{n+1}{n+3}\right)^{n+1}\cdot\frac{3n+5}{n+3}\end{aligned}$$

$$\begin{aligned}F(\mu) < \frac{1}{2} &\iff 2\cdot\frac{3n+5}{n+3} < \left(\frac{n+3}{n+1}\right)^{n+1} \\ &\iff 6 - \frac{8}{n+3} < \left(1+\frac{2}{n+1}\right)^{n+1}\end{aligned}$$

$$\left(1+\frac{2}{n+1}\right)^{n+1} \geq 1 + 2 + \binom{n+1}{2}\left(\frac{2}{n+1}\right)^2 = 3 + \frac{2n}{n+1}$$

$$3 + \frac{2n}{n+1} - \left(6 - \frac{8}{n+3}\right) = \frac{(n-1)(n+5)}{(n+1)(n+3)} > 0$$

$$\because F(median) = \frac{1}{2}$$

$$\therefore \mu < median$$

(iii)

$$\begin{aligned} f'(x) &= k\left(nx^{n-1}(1-x) - x^n\right) \\ &= kx^{n-1}\left(n - (n+1)x\right) \end{aligned}$$

$$f'(x) = 0 \implies x = \frac{n}{n+1} = Mode$$

$$\begin{aligned} F(M) &= \left(\frac{n}{n+1}\right)^{n+1}\left(n+2-(n+1)\frac{n}{n+1}\right) \\ &= 2\left(\frac{n}{n+1}\right)^{n+1} \\ &= \frac{2}{\left(1+\frac{1}{n}\right)^{n+1}} \end{aligned}$$

$$\because \left(1+\frac{1}{x}\right)^{x+1} \text{ is decreasing}$$

$$\therefore n > 1 \implies \left(1+\frac{1}{n}\right)^{n+1} < \left(1+\frac{1}{1}\right)^2 = 4$$

$$F(Mode) > \frac{1}{2}$$

$$\therefore Mode > Median$$

This solution was submitted by **Jon**
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